

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

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Date: 4/30/2025

Compounds: Amoxicillin, Paracetamol, Methamphetamine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxethace

Formula: C₂₀H₃₂N₂O₆

Weight: 396.48

Drug-likeness: 0.65

Activity:

Antibacterial/Analgesic/Stimulant

Target Analysis

Bacterial Peptidoglycan Synthase

Binding Affinity: -8.5

Critical for antibacterial activity in treating infections

Cyclooxygenase (COX)

Binding Affinity: -7

Important for reducing pain and fever symptoms

DAT and VMAT2

Binding Affinity: -9

Increased dopamine and norepinephrine release

Toxicity Profile

Risk Level: High

Affected Systems: Liver, Heart, Brain

Mitigation Suggestions:

- Regular liver function tests
- Monitor cardiovascular health
- Avoid CNS stimulants

Pharmacokinetics

Absorption:

Moderate oral bioavailability

Distribution:

Wide tissue distribution, crosses blood-brain barrier

Metabolism:

Hepatic metabolism via cytochrome P450 enzymes

Excretion:

Renal excretion

Half-life: 12-15 hours

Detailed Analysis

The synthetic compound Amoxethace combines structural elements of amoxicillin, methamphetamine, and paracetamol, leading to a multi-functional drug with antibacterial, analgesic, and stimulant properties. The integration of a beta-lactam moiety from amoxicillin provides effective bacterial cell wall inhibition, while the combination of an arylacetamide moiety from paracetamol grants notable COX inhibition and analgesia. Additional phenylethylamine derived from methamphetamine confers potent effects on the monoamine neurotransmitter system, promoting dopamine and norepinephrine release, hence offering stimulant action. This unique pharmacological profile targets bacterial infections, pain, and neurological conditions in unison, yet it poses significant risks—primarily hepatotoxicity, cardiovascular stress, and CNS overactivity. The phased clinical development will focus initially on establishing safety through robust pharmacokinetic profiling, reducing risks before advancing to efficacy trials.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>